

AUTONOMOUS VEHICLE RESEARCH

# Autonomous Vehicle Research Department.

We connect driving data, model development, simulation and vehicle experiments to advance autonomous driving.

Efficient learning, reasoning and decision-making in complex traffic.

PHENIKAAX / AVR / VIETNAM



Real vehicles. Driving data. Research questions.

CURRENT STAGE

E2E: on-vehicle proof of concept. VLA, reconstruction and simulation: under development.

## RESEARCH PLATFORM

# From driving data to vehicle evaluation



Synchronized camera views, annotations and a bird's-eye view.

## 01 Data collection & curation

Multi-sensor capture, calibration, timestamp alignment, semi-automatic annotation and visualization.

## 02 Driving models

Perception, prediction and planning, with temporal context and vision-language-action models.

## 03 Real-time integration

PhenikaaX Drive Platform supports efficient communication and on-vehicle execution.

## 04 Simulation & evaluation

HD-map-based and camera-based simulation development, closed-loop evaluation and controlled tests.

## RESEARCH DIRECTIONS / 01-04

# Learning, reconstruction and planning

Selected ongoing and planned research. Methods and priorities remain open for discussion.

## 01 Temporal learning & asynchronous inference

Understanding how differences in data sampling, training sequences and inference timing affect end-to-end driving. We explore temporal representations and asynchronous execution for consistent decisions under real-time constraints.

## 02 Efficient end-to-end architectures

Improving driving performance within on-vehicle compute and memory budgets. We study temporal and multi-camera representations, efficient architectures, and the trade-offs between driving quality, inference speed and sensor-to-action latency.

## 03 4D reconstruction & simulation

Improving geometry, appearance and motion in reconstructed driving scenes, including small objects, occlusions and dynamic road users. We investigate how detail and temporal consistency support more faithful simulation and driving evaluation.

## 04 Diverse trajectories & policy learning

Generating varied, dynamically feasible driving trajectories and improving policies through reinforcement learning. We explore learning from driving data, reward design and evaluation methods that reduce dependence on costly simulation.

## RESEARCH DIRECTIONS / 05-08

# Control, reasoning, data and evaluation

Selected ongoing and planned research. Methods and priorities remain open for discussion.

## 05 Learning-assisted vehicle control

Investigating reinforcement learning to adapt model predictive control weights to changing driving conditions. Evaluation considers tracking accuracy, comfort, robustness and real-time computation within vehicle dynamics and operating constraints.

## 06 Vision-language-action driving models

Connecting visual observations, contextual reasoning and driving actions. We study temporal grounding, efficient inference and how reasoning can contribute to end-to-end driving decisions in complex traffic.

## 07 Driving datasets & unified benchmarks

Developing a research dataset covering long-tail situations in motorcycle-rich Asian traffic. We aim to connect perception, motion, reasoning and action through consistent annotations, data selection and benchmarks, with a path toward public release.

## 08 Closed-loop evaluation & generalization

Evaluating model behaviour beyond recorded-data metrics. We study scenario coverage, failure analysis and generalization across traffic conditions, using simulation and controlled vehicle experiments to assess the consequences of driving decisions.

DEMONSTRATIONS

See the work in motion.

Select a video button or media title to open the original video.



End-to-end driving

On-vehicle E2E proof of concept.

Watch video



Data visualization & workflow

Synchronized data and annotations.

Watch video



Ocean Park AV

Autonomous vehicle demonstration.

Watch video

IN THE MEDIA



VTV feature >

Hanoi TV feature >

Vehicle and project coverage

PEOPLE & CONTACT

# Research and engineering team.

We welcome research collaborations, advisory partnerships and student research opportunities. Our team contributes implementation, experiments, driving data and vehicle integration, with project scope agreed together.

**Lê Anh Sơn**  
COMPANY DIRECTOR  
  
PhenikaaX

**Vũ Thành Đạt**  
DEPARTMENT HEAD / PROJECT LEAD  
  
Autonomous Vehicle Research Department (AVR)

**Lý Hoàng Hiệp**  
PROJECT COORDINATOR  
  
Autonomous vehicle project

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PROJECT COORDINATOR  
  
Autonomous vehicle project

CORE RESEARCH & DEVELOPMENT

E2E / Perception · Reasoning / Planning · Sensing / System  
Navigation · Localization · Control

COMPANY CONTACT

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RESEARCH WEBSITE

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